

NAD+

Nicotinamide Adenine Dinucleotide | 500mg & 1,000mg x 10 Vials

SUBCUTANEOUS INJECTION GUIDE | Self-Administration Protocol

No IV Access Required | At-Home Protocol | Daily or 3-5x/Week

Purity: 99.8% | Endotoxin-Free | Research Use Only

This guide covers the Subcutaneous (SubQ) injection protocol for NAD+. SubQ delivery achieves sustained plasma NAD+ elevation without the rate-dependent side effects of IV (nausea, flushing, chest tightness). While peak plasma levels are lower than IV, SubQ NAD+ is absorbed steadily over 30-90 minutes, does not require clinical IV access, and can be self-administered at home. Bioavailability via SubQ is estimated at 85-90% of injected dose.

SubQ vs IV -- Why SubQ is Preferred for At-Home NAD+ Research

Aspect	SubQ Injection	IV Drip
Requires IV access	NO	YES
Clinical setting needed	NO	YES
Side effects (nausea/flush)	MINIMAL	COMMON
Rate-related chest tightness	NONE	COMMON
Self-administration	YES -- easy	Requires training
Bioavailability	~85-90%	100%
Session duration	1-5 min	30-60+ min

Parameter	Specification
Vial Sizes	500mg x 10 1,000mg x 10 Purity 99.8%
Diluent	10mL 0.9% Sodium Chloride (saline) per vial Do NOT use BAC Water for NAD+
Concentrations	500mg vial: 10mL saline -> 50mg/mL (50,000 mcg/mL) 1,000mg vial: 10mL saline -> 100mg/mL
SubQ Dose Range	25-100mg per injection Start at 25mg Build to 50-100mg over 1-2 weeks
Needle	29-31G x 5/16 inch (8mm) insulin syringe 1mL barrel
Injection Site	Abdomen (2+ inches from navel) preferred Rotate 8-point map
Frequency	3-5x per week Daily acceptable No fasting required
Storage (CRITICAL)	2-8 deg C unreconstituted. Reconstituted: USE WITHIN 24 HOURS -- NAD+ oxidises rapidly. Never freeze.

NAD+ MECHANISM OF ACTION | WHY SUBCUTANEOUS WORKS

NAD+ injected subcutaneously is absorbed via dermal capillaries and lymphatics into systemic circulation. Unlike oral NMN or NR (which require hepatic conversion), injectable NAD+ enters cells directly as the intact coenzyme. SubQ absorption is slower than IV, producing a sustained plasma peak over 30-90 minutes rather than an immediate spike -- this slower delivery is actually preferred for tolerability and cellular uptake.

Pathway	Mechanism	Research Benefit
Sirtuin Activation	NAD+ is obligate cofactor for SIRT1-7 (class III HDACs); SIRT1 deacetylates PGC-1alpha (mitochondrial biogenesis), p53, NF-kB; SIRT3 activates mitochondrial antioxidant SOD2; SIRT6 promotes DNA repair; SubQ NAD+ provides sustained cofactor availability	Anti-aging; mitochondrial function; metabolic regulation; DNA repair
PARP DNA Repair	PARP1/2 consume NAD+ for poly-ADP-ribosylation at DNA break sites; high DNA damage (UV, oxidative stress, aging) depletes NAD+; restoring NAD+ via SubQ injection enhances DNA repair capacity	Genomic stability; cancer prevention research; aging
Mitochondrial ATP Synthesis	NAD+/NADH ratio drives electrons through Complex I -> ETC; restoring NAD+ improves Complex I activity and membrane potential; SubQ delivery produces gradual sustained elevation unlike IV spike	Energy metabolism; chronic fatigue; mitochondrial disease
SubQ vs IV Pharmacokinetics	SubQ: gradual absorption 30-90 min, bioavailability ~85-90%, no rate-dependent side effects, no IV access required; IV: immediate peak, 100% bioavailability, nausea/flushing common, requires clinical setting; both achieve meaningful NAD+ elevation	SubQ preferred for: home use, frequent dosing, tolerability-sensitive subjects

Why SubQ Injection Outperforms Oral Supplementation

Route	Bioavailability	Peak Time	Side Effects	Convenience	Home Use
SubQ Injection (This Protocol)	~85-90%	30-90 min (gradual)	Minimal	Moderate	YES
IV Drip	100%	During infusion	Common (rate-dependent)	Low	NO (clinic required)
Oral NMN/NR (precursor)	~5-25% (variable)	2-4 hours (indirect)	Minimal	High	YES

RECONSTITUTION & DOSE CALCULATION | SubQ PROTOCOL

CRITICAL: Use 0.9% Sodium Chloride (saline) -- NOT BAC Water. Use reconstituted NAD+ within 24 hours -- NAD+ oxidises rapidly in solution. Prepare fresh each session. Never store reconstituted NAD+ overnight.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather supplies	NAD+ vial, 0.9% NaCl saline (10mL), insulin syringe (29-31G x 5/16in), drawing needle (18-20G optional), alcohol swabs, sharps container.
2	Draw 10mL saline	Using a 10mL syringe or insulin syringe in multiple draws, add 10mL of 0.9% NaCl to the NAD+ vial.
3	Inject saline into vial	Inject saline slowly against the GLASS WALL of the vial. Never directly onto the powder. Both vial sizes use exactly 10mL.
4	Swirl gently until clear	Roll vial between palms 20-30 seconds. NAD+ dissolves rapidly. Solution should be clear to very slightly yellow. NEVER shake.
5	Draw SubQ dose immediately	Using a 29-31G insulin syringe, draw your dose per the table below. Inject immediately or within 2 hours. Discard all unused solution at the end of the session.

SubQ Dose Calculation -- 500mg Vial (10mL saline -> 50mg/mL)



Use 29-31G insulin syringe | 1mL barrel | SubQ into abdomen

500mg vial (10mL saline -> 50mg/mL)

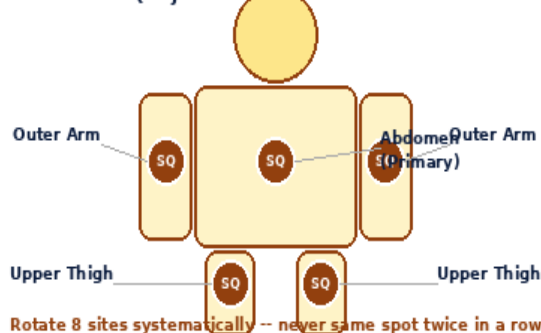
Dose (mg)	25mg	50mg	100mg	200mg
Volume (mL)	0.500	1.000	2.000	4.000
Syringe Units	50U	100U	200U	400U

1,000mg vial (10mL saline -> 100mg/mL)

Dose (mg)	25mg	50mg	100mg	200mg
Volume (mL)	0.250	0.500	1.000	2.000
Syringe Units	25U	50U	100U	200U

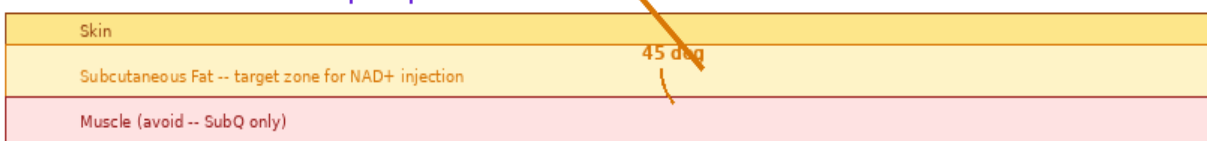
INJECTION TECHNIQUE, SCHEDULE & PROTOCOLS

NAD+ SubQ Injection Sites -- Abdomen Preferred



45-Degree SubQ Injection Angle

Pinch skin



Step	Action	Detail
1	Wash hands	Wash with soap and warm water 20 seconds. Dry completely.
2	Reconstitute & draw	Add 10mL saline to vial. Swirl gently until clear. Draw exact dose into 29-31G insulin syringe. Expel air bubbles.
3	Clean & inject	Wipe site with alcohol swab. Dry 10-15 sec. Pinch 2-3cm skin. Insert needle at 45 degrees. Release pinch. Inject slowly over 10-20 seconds.
5	Withdraw and press	Remove needle at same angle. Press gently with dry swab 10-15 seconds. Do NOT rub.
6	Discard & rotate	Needle immediately into sharps container. Record site used. Move to next rotation point for next session.

Recommended SubQ Dosing Schedules -- Choose One

3x/week (Beginner)	Mon INJECT	Tue	Wed INJECT	Thu	Fri INJECT	Sat	Sun
5x/week (Standard)	Mon INJECT	Tue INJECT	Wed INJECT	Thu INJECT	Fri INJECT	Sat	Sun

Next empty days -- when leveling schedule start 2x week before moving to 5x

SubQ Dosing Protocols

Protocol	Dose per Injection	Frequency	Vial Used	Notes
Beginner Start	25mg	3x/week	500mg vial (50 Units)	Week 1-2. Assess tolerance. Mild flush/warmth is normal.
Standard Protocol	50mg	5x/week	500mg vial (100 Units/1mL)	Most common research dose. Good tolerability profile.

Side Effects -- SubQ Route vs IV

Side Effect	SubQ Likelihood	Management
Mild warmth at injection site	Common	Normal vasodilatory response. Resolves within 30 min. Not concerning.
Mild flushing (face/chest)	Occasional	Far less than IV. Dose-related. Start at 25mg if sensitive.
Mild fatigue post-injection	Occasional	NAD+ drives cellular metabolic work. Rest after injection if needed.
Injection site redness/itch	Occasional	Rotate sites. Use fresh needle. Warm solution to room temp.
Nausea	Rare (vs common IV)	Major advantage of SubQ over IV. Start low (25mg) if history of nausea.
Chest tightness	Absent (vs common IV)	Rate-dependent IV side effect -- does NOT occur with SubQ.
Muscle cramps (IV)	Absent (SubQ)	IV side effect eliminated by SubQ route.

Golden Rules -- NAD+ SubQ Protocol

1. Use 0.9% saline (NOT BAC Water) to reconstitute NAD+. Both vial sizes: add 10mL saline.
2. CRITICAL: Use reconstituted NAD+ within 24 hours -- it oxidises rapidly. Prepare fresh each session.
3. Start at 25mg (25-50 Units on syringe) to assess tolerance before escalating.
4. SubQ eliminates IV side effects: no chest tightness, minimal nausea, no clinical setting needed.
5. Inject slowly over 10-20 seconds. Use a fresh 29-31G needle every injection.
6. Allow each site 7+ days rest before reuse.
7. Standard protocol: 50mg SubQ 5x/week. Build from 25mg over 1-2 weeks. Store 2-8 deg C.

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